

Chapter 1 Basic Ideas in I

1.1 Basic Concepts

Random experiments: ① repeatable ② all possible outcomes can not be predicted exactly one outcome each time, known risk
随机试验

Sample space: $\Omega = \{\text{All possible cases}\}$

样本空间:

Sample points ω
样本点

Random events: A

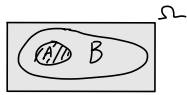
随机事件

Probability $P(A)$
概率

$P(A) \geq 0$ Non-negativity
 $P(\Omega) = 1$ Normalization
 $P(\bigcup_{i=1}^n A_i) = \sum_{i=1}^n P(A_i)$ Additivity

Operations: ① Finite-additivity ② $P(\emptyset) = 0$
 $P(\bigcup_{i=1}^n A_i) = \sum_{i=1}^n P(A_i)$

① Inclusion



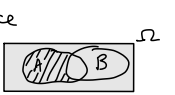
② Sum/Union



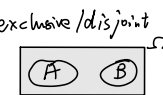
③ Product/Intersection



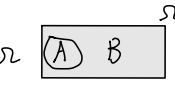
④ Difference



⑤ Mutually exclusive/disjoint



⑥ Complement



Rules

1. Commutative laws

$$A \cup B = B \cup A, A \cap B = B \cap A$$

2. Associative laws

$$A \cup (B \cap C) = (A \cup B) \cap C$$

$$A \cap (B \cup C) = (A \cap B) \cup C$$

3. Distributive laws

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

4. De Morgan's laws

$$\overline{\bigcup_{i=1}^n A_i} = \bigcap_{i=1}^n \overline{A_i}, \quad \overline{\bigcap_{i=1}^n A_i} = \bigcup_{i=1}^n \overline{A_i}$$

Probability properties

补 ① $P(\bar{A}) = 1 - P(A)$

界 ② $0 \leq P(A) \leq 1$

单 ③ $A \subseteq B \Rightarrow P(A) \leq P(B), P(B-A) = P(B) - P(A)$

加 ④ $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

容斥 ⑤ $P(A_1 \cup A_2 \cup A_3 \dots \cup A_n) = P(A_1) + P(A_2) + P(A_3) + \dots + P(A_n) - P(A_1 A_2) - P(A_1 A_3) - \dots + (-1)^{n-1} P(A_1 A_2 \dots A_n)$

$$= \sum_{i=1}^n P(A_i) - \sum_{1 \leq i < j \leq n} P(A_i A_j) + \sum_{1 \leq i < j < k \leq n} P(A_i A_j A_k) - \dots + (-1)^{n-1} P(A_1 A_2 \dots A_n)$$

1.2 Computing Probabilities

古典概型

$$P(A) = \frac{\# \text{ of } A}{\# \text{ of } \Omega}$$

Notice the change of the sample space

e.g. n students
n gifts from each one

$$P(A) = \{ \text{At least 1 student got his/her own gifts} \} = ?$$

Solution: Break down

$$A = A_1 \cup A_2 \cup A_3 \cup \dots \cup A_n$$

n-th student got his/her own gifts

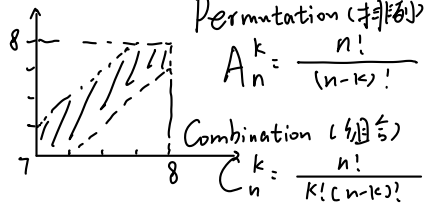
$$P(A) = P(A_1 \cup A_2 \cup \dots \cup A_n) = \sum_{i=1}^n P(A_i) - \sum_{i < j} P(A_i A_j) + \dots + (-1)^{n-1} P(A_1 A_2 \dots A_n)$$

凡可排列

$$P(A) = \frac{L/A/V \text{ of } A}{L/A/V \text{ of } \Omega}$$

e.g. A, B come in 7:00 - 8:00 and stay for 15 min

$$P(A, B \text{ meet}) = ?$$



Permutation (排列)

$$A_n^k = \frac{n!}{(n-k)!}$$

Combination (组合)

$$C_n^k = \frac{n!}{k!(n-k)!}$$

不放回抽样

1.3 Conditional Probability and Independence

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

B updates the sample space, also updates our belief on P(A)

① Multiplication law 乘法定律

$$P(A \cap B) = P(A|B)P(B) \text{ if } P(B) > 0$$

Chain rule:

$$P(A_1 A_2 A_3 \dots A_n) = P(A_1 | (A_2 \dots A_n) A_n) = P(A_n | A_1 \dots A_{n-1}) P(A_1 | A_2 \dots A_{n-1}) P(A_2 | A_1) P(A_1)$$

② Law of Total probability 全概率定律

$$P(A) = P(A \cap \Omega) = P(A \cap B) + P(A \cap \bar{B}) = P(A|B)P(B) + P(A|\bar{B})P(\bar{B})$$

$$= P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + \dots + P(A|B_n)P(B_n)$$

③ Bayes' theorem

$$B_1 \cap B_2 \cap \dots \cap B_n = \Omega$$

Let $B_1, B_2, \dots, B_n$ be partition of the sample space

$$P(B_i|A) = \frac{P(A|B_i)P(B_i)}{P(A)} = \frac{P(B_i)P(A|B_i)}{\sum_{j=1}^n P(B_j)P(A|B_j)}$$

$P(B_i|A)$ posterior probability 后验概率

$P(B_i)$ Prior probability 先验概率

④ Independence

$$P(A|B) = P(A), P(B|A) = P(B)$$

$$P(A \cap B) = P(A|B)P(B) = P(A)P(B)$$

The occurrence of A does not affect the probability of B

$$P(A_1 A_2 \dots A_n) = P(A_1)P(A_2) \dots P(A_n)$$

if $A_1, \dots, A_n$ are independent

⑤ Conditional Independence 条件独立

Let $A_1, \dots, A_n$ be random events

$A_1, \dots, A_n$ are conditioned on B , if for $\forall$ subset of the events $A_{i_1}, A_{i_2}, \dots, A_{i_k}$ ($1 \leq i_1 < \dots < i_k \leq n, k=2, \dots, n$)

$$P(A_{i_1} A_{i_2} \dots A_{i_k} | B) = P(A_{i_1} | B) P(A_{i_2} | B) \dots P(A_{i_k} | B)$$

e.g. N words in an email, W_i is the i-th word in the email, S denotes that the mail is the spam.

$$P(S | W_1, \dots, W_N) = \frac{P(W_1, \dots, W_N | S) P(S)}{P(W_1, \dots, W_N | S) P(S) + P(W_1, \dots, W_N | \bar{S}) P(\bar{S})}$$

(Assuming conditional independence of word appearing)

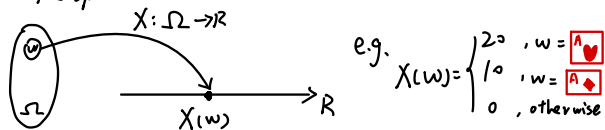
$$\Rightarrow P(W_1, \dots, W_N | S) = P(W_1 | S) P(W_2 | S) \dots P(W_N | S)$$

Chapter 2 Random Variables and Distribution

2.1 Introduction

① Definition

random variable is a real-valued function defined on the sample space



Probability distribution (概率分布):

- Examining all values r.v can take.
- Find the probability associated with each value.

② Based on the values a r.v can take, 2 types of variable:

Discrete random variables: Finite values: e.g. number of customers

Continuous random variables: Continuous value: time of meeting

- Discrete r.v: Probability mass function PMF
- Continuous r.v: Probability density function PDF
- both r.v: cumulative distribution function CDF

(i) PMF

let X be a discrete r.v, $S = \{a_1, a_2, \dots\}$

S is the support (支撑集) of X

The PMF of X is $p: S \rightarrow [0, 1]$

$$p(a_i) = p_i = P(X=a_i), i=1, 2, \dots$$

$$\Rightarrow P(a_i) \geq 0, \sum_i P(a_i) = 1$$

Value	a_1	a_2	$\dots$	a_i
Probability	p_1	p_2	$\dots$	p_i

(ii) PDF

let x be a continuous r.v

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

$f(x) \geq 0, \int_{-\infty}^{\infty} f(x) dx = 1$
 $\Rightarrow$ reflect the degree to which P is concentrated near x
 $\Rightarrow \int_x^x f(x) dx = 0 \Rightarrow$ Not mean $P(X=x)$

(iii) CDF

let x be a r.v

$$F(x) = P(X \leq x) \begin{cases} F(x) = \sum_{a_i \leq x} p(a_i) \\ F(x) = \int_{-\infty}^x f(u) du \\ f(x) = F'(x) \end{cases}$$

$\Rightarrow$ Non-decreasing right-continuous

$$\text{Max: } \lim_{x \rightarrow \infty} F(x) = 1$$

$$\text{Min: } \lim_{x \rightarrow -\infty} F(x) = 0$$

$$P(a < x \leq b) = F(b) - F(a)$$

③ Expectation and Variance 期望和方差

$$E(X) = \begin{cases} \sum_{k=1}^{\infty} x_k P(X=x_k) = \sum_{k=1}^{\infty} x_k p_k & \text{Discrete r.v.} \\ \int_{-\infty}^{\infty} x f(x) dx & \text{Continuous r.v.} \end{cases}$$

$$\text{Var}(X) = E(X - E(X))^2 = E(X^2) - (E(X))^2$$

SD 标准差 $\triangleq \sqrt{\text{Var}(X)}$

Discrete PMF: $P(X_k) = p_k = P(X=x_k), \text{Var}(X) = \sum_{k=1}^{\infty} (x_k - E(X))^2 \cdot p_k$

Continuous PDF: $\text{Var}(X) = \int_{-\infty}^{\infty} (x - E(X))^2 f(x) dx$

$E(aX + b) = aE(X) + b$

$\text{Var}(aX + b) = a^2 \text{Var}(X)$

$\text{Var}(X) = E(X^2) - (E(X))^2$

2.2 Common Discrete Distribution

① Bernoulli distribution 伯努利分布

$X \sim \text{Bernoulli}(p)$, if X takes 0 and 1,

$$P(X=1) = p, P(X=0) = 1-p$$

$$\text{PMF: } P(X) = p^x (1-p)^{1-x}, x=0, 1$$

$$E(X) = p, \text{Var}(X) = p(1-p)$$

② Binomial Distribution 二项分布

n -fold Bernoulli Trial

$p(A)$ in each trial same

$p(A$ happens k times in n trials)

$\downarrow$ Binomial distribution $X \sim \text{Bin}(n, p)$ n-fold bernoulli trial

$$\text{PMF: } P(X) = \binom{n}{x} p^x (1-p)^{n-x} = C_n^x p^x (1-p)^{n-x}$$

$$E(X) = np, \text{Var}(X) = np(1-p)$$

③ Geometric Distribution 几何分布

Bernoulli trial repeated, until A happens

X is the number of trial required

$X \sim \text{Geo}(p)$

$$\text{PMF: } P(X) = p(1-p)^{x-1}, x=1, 2, \dots$$

$$E(X) = \frac{1}{p}, \text{Var}(X) = \frac{1-p}{p^2}$$

Only discrete distribution **MEMORYLESS** (无记忆性) **PROPERTY**

$$\Rightarrow X \sim \text{Geo}(p) \Rightarrow P(X > m+n | X > m) = P(X > n)$$

④ Poisson Distribution 泊松分布

* Events happen in a fixed time/space

if they occur with a constant rate and independently

$X \sim \text{Poisson}(\lambda)$

$$\text{PMF: } P(X) = P(X=x) = \frac{\lambda^x}{x!} e^{-\lambda}, x=0, 1, 2, \dots$$

$$E(X) = \lambda, \text{Var}(X) = \lambda$$

$$\text{Poisson theorem: } \lim_{n \rightarrow \infty} P(X=x) = \frac{\lambda^x}{x!} e^{-\lambda}$$

$$\text{Poisson } (\lambda) \xrightarrow{\text{Approximate}} \text{Binomial } (n, p) \text{ when } n > 10, p < 0.5 \Rightarrow \lambda \approx np$$

2.3 Common Continuous Distribution

① Uniform Distribution

$X \sim U(a, b)$

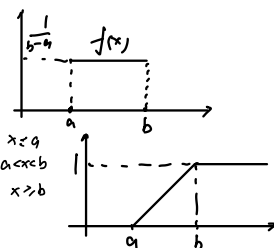
$$\text{PDF: } f(x) = \begin{cases} \frac{1}{b-a}, & \text{if } a < x < b \\ 0, & \text{otherwise} \end{cases}$$

$$\text{CDF: } F(x) = P(X \leq x) = \int_{-\infty}^x f(u) du = \begin{cases} 0, & x < a \\ \frac{x-a}{b-a}, & a < x < b \\ 1, & x > b \end{cases}$$

$\Rightarrow$ for $\forall (c, c+L) \in (a, b)$

$$P(c < X < c+L) = \frac{L}{b-a}$$

$$E(X) = \frac{a+b}{2}, \text{Var}(X) = \frac{(b-a)^2}{12}$$



⊕ Exponential Distribution

$$X \sim \text{Exp}(\lambda)$$

$$\text{PDF} = f(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

$$\text{CDF} = F(x) = P(X \leq x) = \int_{-\infty}^x f(u) du = \begin{cases} 1 - e^{-\lambda x}, & x \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

$$E(X) = \frac{1}{\lambda}, \text{Var}(X) = \frac{1}{\lambda^2}$$

Only continuous distribution **MEMORYLESS**
(无记忆性) **PROPERTY**

if $X \sim \text{Exp}(\lambda)$
 $P(X > s+t | X > s) = P(X > t)$

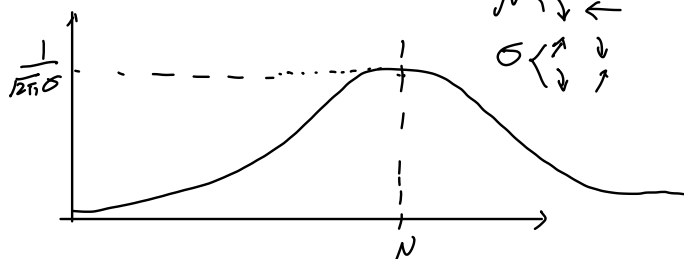
⊕ Normal Distribution

$$X \sim N(\mu, \sigma^2)$$

$$\text{PDF} = f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, -\infty < x < \infty$$

* $\mu=0, \sigma=1 \Rightarrow$ Standard normal distribution

$$\text{PDF} = \phi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$$



* $X \sim N(\mu, \sigma^2), Z = \frac{X-\mu}{\sigma} \rightarrow Z \sim N(0, 1)$

$$E(X) = \mu, \text{Var}(X) = \sigma^2$$

* if $X \sim N(\mu, \sigma^2), Z = \frac{X-\mu}{\sigma}, Z \sim N(0, 1)$

$$P(X \leq x) = P(Z \leq \frac{x-\mu}{\sigma}) = \Phi(\frac{x-\mu}{\sigma})$$

$$\text{if } Z \rightarrow X = \mu + (Z \times \sigma) = \text{standard deviation of } X$$

2.4 Transformation of Random Variables

Discrete to Discrete:

X with PMF $P_X(x)$, $Y = g(X)$, $P_Y(y) = \sum_{x: g(x)=y} P_X(x)$

Continuous to Discrete:

X with PDF $f_X(x)$, $Y = g(x)$, $P_Y(y) = \int_{x: g(x)=y} f_X(x) dx$

Continuous to Continuous:

X with PDF $f_X(x)$, $Y = g(x)$

↓
If $g(x)$ is 严格单调函数, and $h(y) = g^{-1}(y)$

↓
 $f_Y(y) = \begin{cases} |h'(y)| f_X(h(y)), & \text{where } h(y) \text{ is defined} \\ 0, & \text{otherwise} \end{cases}$

* If X continuous, CDF $F_X(x)$ & $F_X'(x)$ exists

$\Rightarrow Y = F_X(x) \sim \text{Uni}(0,1)$

⊕ If $F_X(x)$ is CDF, $F_X^{-1}(x)$ exists, for $X = F_X^{-1}(U)$, $X \sim F_X(x)$
逆变换采样

Expectation: $Y = g(X)$, $E(Y) = ?$

⊕ If X discrete, $E(Y) = \sum_{k=1}^{\infty} g(x_k) p_k$

⊕ If X continuous, $E(Y) = \int_{-\infty}^{\infty} g(x) f(x) dx$

$E(g(x))$ and $g(E(x))$, for $g(x)$

⊕ If Linear, $E(g(x)) = g(E(x))$

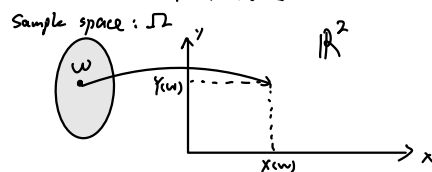
⊕ If convex, $E(g(x)) \geq g(E(x))$

⊕ If concave, $E(g(x)) \leq g(E(x))$

Chapter 3 Joint Distribution

3.1 Random Vector and joint distribution

(X, Y) is a random vector, if $\omega \in \Omega \rightarrow (X(\omega), Y(\omega))$
随机向量



Joint CDF: $F(x, y) = P(X \leq x, Y \leq y), \forall x, y \in \mathbb{R}$

A: $X \leq x, B: Y \leq y, F(x, y) = P(A \cap B)$

Drops: $F(+\infty, +\infty) = 1, F(-\infty, -\infty) = 0$

$F(-\infty, y) = 0, F(x, -\infty) = 0$

Marginal CDF: $F_X(x)$ is joint CDF

$F_X(x)$ is the CDF of X without considering Y

$F_Y(y)$ is the CDF of Y without considering X

$\Rightarrow F_X(x) = F(x, \infty), F_Y(y) = F(\infty, y)$

F_X is called marginal CDF of X

F_Y is called marginal CDF of Y

Joint and Marginal PMF

$(X, Y), S_X = \{x_1, x_2, \dots\}, S_Y = \{y_1, y_2, \dots\}$ * $P_{ij} \geq 0, \sum_i \sum_j P_{ij} = 1$
 $P(X=x_i, Y=y_j) = P_{ij}, i, j = 1, 2, \dots$

marginal PMF: $P_X(x_i) = \sum_{j=1}^{\infty} P_{ij} \triangleq p_i, i = 1, 2, \dots$

$X \backslash Y$	y_1	y_2	y_3	...
x_1	p_{11}	p_{12}	p_{13}	...
x_2	p_{21}	p_{22}	p_{23}	...
x_3	p_{31}	p_{32}	p_{33}	...
...	...	...	...	...

$P_Y(y_j) = \sum_{i=1}^{\infty} P_{ij} \triangleq p_j, j = 1, 2, \dots$

If $P(X=x_i, Y=y_j) = P_X(x_i) P_Y(y_j)$ independent

Joint and Marginal PDF:

$(X, Y) f(x, y) \geq 0, \forall (x, y) \in \mathbb{R}^2, \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$
for $\forall D \in \mathbb{R}^2$

$P((X, Y) \in D) = \iint_{(x, y) \in D} f(x, y) dx dy$

CDF: $F(x, y) = \int_{-\infty}^y \int_{-\infty}^x f(u, v) du dv, f(x, y) = \frac{\partial^2}{\partial x \partial y} F(x, y)$

Marginal:

$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy, f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx$

Conditional PMF/PDF

① For a discrete (X, Y) , PMF $p(x, y)$
the conditional PMF of X given $Y=y$ is

$$P_{X|Y}(x|y) \triangleq P(X=x|Y=y) = \frac{P(x, y)}{P_Y(y)}, \text{ for all value of } y \text{ that } P_Y(y) > 0$$

② For a continuous (X, Y) , PDF $f(x, y)$
the conditional PDF of X given $Y=y$ is

$$f_{X|Y}(x|y) \triangleq \frac{f(x, y)}{f_Y(y)} \text{ for all values of } y \text{ that } f_Y(y) > 0$$

* If $f_{X|Y}(x|y) \neq f_X(x)$, X not independent of Y

3.2 Relation between Two Random Variables

Independence of R.V

Continuous Case:

$$\Rightarrow f_{X|Y}(x|y) = f_X(x) \Rightarrow f(x, y) = f_{X|Y}(x|y) f_Y(y) = f_X(x) f_Y(y)$$

$$\Rightarrow f_{Y|X}(y|x) = f_Y(y) f_X(x)$$

$$\Rightarrow F(x, y) = \int_{-\infty}^y \int_{-\infty}^x f(u, v) du dv = \int_{-\infty}^x f_X(u) du \int_{-\infty}^y f_Y(v) dv = F_X(x) F_Y(y)$$

$\Rightarrow$ Let $F(x_1, x_2, \dots, x_n)$ be CDF of $\{X_1, \dots, X_n\}$

$F_{X_i}(x_i)$ be the marginal CDF of $X_i \Rightarrow$

$$F(x_1, x_2, \dots, x_n) = F_{X_1}(x_1) F_{X_2}(x_2) \dots F_{X_n}(x_n)$$

$X_1, \dots, X_n$ are mutually independent

$$\left\{ \begin{array}{l} \text{discrete } p(x_1, \dots, x_n) = P(X_1=x_1, \dots, X_n=x_n) = p_{X_1}(x_1) p_{X_2}(x_2) \dots p_{X_n}(x_n) \\ \text{continuous } f(x_1, \dots, x_n) = f_{X_1}(x_1) f_{X_2}(x_2) \dots f_{X_n}(x_n) \end{array} \right.$$

Jacobian Determinant

$$\begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}$$

Covariance 协方差 (Unit matters)

properties: X, Y independent r.v.s, for any function g, h

$$\textcircled{1} E(g(X)h(Y)) = E(g(X))E(h(Y))$$

$$\textcircled{2} \text{Var}(g(X) \pm h(Y)) = \text{Var}(g(X)) + \text{Var}(h(Y))$$

Def: $\text{Cov}(X, Y) \triangleq E[(X-E(X))(Y-E(Y))] = E(XY) - E(X)E(Y)$

If X, Y independent $\Leftrightarrow \text{Cov}(X, Y) = 0 \Leftrightarrow X, Y$ uncorrelated

props: $\text{Cov}(X, X) = \text{Var}(X)$ $\text{Cov}(aX, bY) = ab \text{Cov}(X, Y)$ 不相关

$$\text{Cov}(X, Y) = \text{Cov}(Y, X) \quad \text{Cov}(X_1 + X_2, Y) = \text{Cov}(X_1, Y) + \text{Cov}(X_2, Y)$$

$$\text{Cov}(X, c) = 0 \quad \text{Cov}(X+a, Y+b) = \text{Cov}(X, Y)$$

$$\text{Cov}(a_1X_1 + \dots + a_nX_n, b_1Y_1 + \dots + b_mY_m) = \sum_{i=1}^n \sum_{j=1}^m a_i b_j \text{Cov}(X_i, Y_j)$$

$$\text{Cov}(aX+bY, cW+dV) = ac \text{Cov}(X, W) + ad \text{Cov}(X, V) + bc \text{Cov}(Y, W) + bd \text{Cov}(Y, V)$$

Unit: $X(\text{in cm}), \tilde{X}(\text{in cm}) \text{Cov}(\tilde{X}, Y) = 100 \text{Cov}(X, Y)$

When X, Y vary in the same direction $\text{Cov}(X, Y) > 0$

When X, Y vary in different direction $\text{Cov}(X, Y) < 0$

Correlation Coefficient 相关系数

$$\text{Cor}(X, Y) = \rho_{XY} \triangleq E\left[\frac{(X-E(X))}{SD(X)} \cdot \frac{(Y-E(Y))}{SD(Y)}\right] = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}}$$

无量纲

Measures the direction and strength of the Linear relationship X, Y

MSE (平均误差) of using Linear function X to approximate $Y, \hat{Y} = a+bX$

$$\text{MSE} = E[(Y-\hat{Y})^2] = E[(Y-a-bX)^2]$$

$$= E(Y^2) + b^2 E(X^2) + a^2 - 2bE(XY) + 2abE(X) - 2aE(Y)$$

minimize MSE w.r.t a, b

$$\begin{cases} \frac{\partial \text{MSE}}{\partial a} = 2a + 2bE(X) - 2E(Y) = 0 \\ \frac{\partial \text{MSE}}{\partial b} = 2bE(X^2) - 2E(XY) + 2aE(X) = 0 \end{cases} \Rightarrow a, b$$

$$\min_{a,b} \text{MSE} = \text{Var}(Y)(1 - \rho_{XY}^2) \geq 0, \rho_{XY}^2 \leq 1$$

$\rho_{XY} \begin{cases} > 0 & \text{positively correlated} \\ = 0 & \text{uncorrelated} \\ < 0 & \text{negatively correlated} \end{cases}$

3.3 Function of Multiple R.V.s

$$(X, Y) \rightarrow g(X, Y)$$

Continuous case: CDF of $Z = g(X, Y) = F_Z(z) = P(Z \leq z) = P(g(X, Y) \leq z)$

$$= \int \int_{g(x, y) \leq z} f(x, y) dx dy = \int_{-\infty}^{\infty} f_2(u) du$$

PDF of Z $Z = X+Y$

$$f_Z(z) = \int_{-\infty}^{\infty} f(z-y, y) dy = \int_{-\infty}^{\infty} f(x, z-x) dx$$

If X, Y independent

$$f_Z(z) = f_X * f_Y \triangleq \int_{-\infty}^{\infty} f_X(z-y) f_Y(y) dy = \int_{-\infty}^{\infty} f_X(x) f_Y(z-x) dx$$

Convolution (卷积)

Discrete Case: support of X and Y are $\{0, 1, 2, \dots\}$, PMF of $Z = X+Y$ is $\{k=0, 1, 2, \dots\}$

$$P(Z=k) = \sum_{i=0}^k P(X=i, Y=k-i) = \sum_{i=0}^k P(X=k-j, Y=j)$$

If X, Y independent

$$P(Z=k) = \sum_{i=0}^k P(X=i) \cdot P(Y=k-i) = \sum_{j=0}^k P(X=k-j) P(Y=j)$$

discrete convolution formula

General result about Sum of Independent Normal R.V.s

$$X \sim N(\mu_1, \sigma_1^2) \quad Y \sim N(\mu_2, \sigma_2^2) \Rightarrow (X+Y) \sim N(\mu_1+\mu_2, \sigma_1^2+\sigma_2^2)$$

If $X_1, \dots, X_n$ independent, $X_i \sim N(\mu_i, \sigma_i^2) \quad i=1, 2, \dots, n$

$$a_1X_1 + a_2X_2 + \dots + a_nX_n \sim N\left(\sum_{i=1}^n a_i\mu_i, \sum_{i=1}^n a_i^2\sigma_i^2\right)$$

CLT 中心极限定理

$S_n = X_1 + \dots + X_n$, $X_1, \dots, X_n$ are independent identical Distribution R.V.s

$$\mu \triangleq E(X_i), \sigma^2 \triangleq \text{Var}(X_i), \text{ Let } S_n = X_1 + \dots + X_n, \bar{X}_n = \frac{S_n}{n}$$

$$Z_n = \frac{S_n - E(S_n)}{\sqrt{\text{Var}(S_n)}} = \frac{S_n - n\mu}{\sqrt{n}\sigma} = \frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \quad (\text{usually } n \geq 30)$$

As $n \rightarrow \infty$, Z_n converges in distribution to a standard normal r.v

$$F_{Z_n}(z) = P(Z_n \leq z) \xrightarrow{n \rightarrow \infty} \phi(z) \text{ for all } z$$

Even if not independent, not identical distribution $\Rightarrow$ still holds

Continuity correction 连续校正

Needed when we approximate a discrete distribution by a continuous one

Expand interval by $\frac{1}{2}$ in each direction

$$P(k \leq X \leq l) = P(k-0.5 \leq X \leq l+0.5) \\ = P\left(\frac{k-0.5-n\mu}{\sqrt{np(1-p)}} \leq \frac{X-n\mu}{\sqrt{np(1-p)}} \leq \frac{l+0.5-n\mu}{\sqrt{np(1-p)}}\right)$$

$$\approx \Phi\left(\frac{l+0.5-n\mu}{\sqrt{np(1-p)}}\right) - \Phi\left(\frac{k-0.5-n\mu}{\sqrt{np(1-p)}}\right)$$

$$\text{Var}(aX \pm bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) \pm 2ab \text{Cov}(X, Y)$$

Chapter 4 Monte Carlo Methods

4.1 Introduction and Theoretical Basis

Law of Large numbers (LLN, 大数定理)

Let $X_1, X_2, \dots$ be a sequence of i.i.d. random variables with $\mu \triangleq E(X_i) < \infty$, $\bar{X}_n = (X_1 + \dots + X_n)/n$
 $\bar{X}_n$ converges in probability (依概率收敛)

$$\bar{X}_n \xrightarrow{P} \mu, \text{ as } n \rightarrow \infty$$

$$\text{for } \forall \epsilon > 0, \lim_{n \rightarrow \infty} P\left(\left|\frac{1}{n} \sum_{i=1}^n X_i - \mu\right| > \epsilon\right) = 0$$

$$\bar{X}_n \overset{\text{approx.}}{\sim} N\left(\mu, \frac{\sigma^2}{n}\right), \text{ as } n \rightarrow \infty, \text{Var}(\bar{X}_n) \rightarrow 0, \text{converges}$$

$\bar{X}_n \xrightarrow{P} \mu$ as $n \rightarrow \infty$, suggesting that probability is the long-term frequency

4.2 Simulation of R.V.s

PRNG 伪随机数生成器

Given a uniform distribution random number generator

$$(1) \text{Bern}(p): X = \begin{cases} 1, & X < p \\ 0, & X \geq p \end{cases} \sim \text{Bern}(p)$$

$$\text{for } u_1, u_2, \dots, u_n, \quad \underline{X}_i = \begin{cases} 1, & u_i < p \\ 0, & u_i \geq p \end{cases}$$

Discrete

$$(2) \text{Poisson}(\lambda) \quad P_k = P(X=k) = \frac{\lambda^k}{k!} e^{-\lambda}, \quad k=0, 1, 2, \dots$$

divide interval $(0, 1)$ to $\frac{[0, p_0]}{1}, \frac{[p_0, p_0+p_1]}{1}, \dots$

Define $\underline{X}_i = k$ if $u_i \in I_k$

$$(3) \text{Exp}(\lambda)$$

$$F(x) = 1 - e^{-\lambda x} \quad F^{-1}(u) = -\frac{1}{\lambda} \ln(1-u)$$

$u_1, u_2, \dots$ from uniform distribution

$$\underline{X}_i = -\frac{1}{\lambda} \ln(1-u_i)$$

$$(4) \text{Gamma}(r, \lambda) \text{ is } r \text{ independent } \text{Exp}(\lambda) \text{ sum}$$

let $X_{11}, X_{12}, \dots, X_{1r}, X_{2r}, \dots, X_{nr}, \dots, X_{nr}$ be rn numbers from $\text{Exp}(\lambda)$

$$\underline{y}_i = X_{i1} + X_{i2} + \dots + X_{ir}, \quad i=1, \dots, n$$

$$f_z(z) = \begin{cases} \frac{\lambda^r}{(r-1)!} z^{r-1} e^{-\lambda z}, & z \geq 0 \\ 0, & \text{other} \end{cases}$$

$$(5) N(\mu, \sigma^2)$$

if X, Y independent

$$E[g(x)h(y)] = E[g(x)]E[h(y)]$$

$$\text{Var}[g(x) \pm h(y)] = \text{Var}[g(x)] + \text{Var}[h(y)]$$

Let $u_1, u_2 \sim [0, 1]$

构造标准正态

$$\begin{cases} Z_1 = \sqrt{-2 \ln(u_1)} \cos(2\pi u_2) \\ Z_2 = \sqrt{-2 \ln(u_1)} \sin(2\pi u_2) \end{cases} \text{ a pair of independent standard normal R.V.s.}$$

$$(X, Y) \rightarrow \max(X, Y), \min(X, Y)$$

Continuous case: $f(x, y)$ be the PDF of (X, Y)

$$F_{\max}(z) = \int_{-\infty}^z \int_{-\infty}^z f(x, y) dx dy, \quad F_{\min}(z) = 1 - \int_{-\infty}^z \int_{-\infty}^z f(x, y) dx dy$$

If x, y independent,

$$F_{\max}(z) = F_X(z)F_Y(z), \quad F_{\min}(z) = 1 - (1 - F_X(z))(1 - F_Y(z))$$

Specially, if $X_1, X_2, \dots, X_n$ i.i.d $F(x)$ with PDF $f(x)$

$$F_{\max}(z) = [F(z)]^n, \quad f_{\max}(z) = n f(z) [F(z)]^{n-1}$$

$$F_{\min}(z) = 1 - [1 - F(z)]^n, \quad f_{\min}(z) = n f(z) [1 - F(z)]^{n-1}$$

Other form, start from CDF and integration

e.g. $Z = X/Y, X, Y \sim \text{Exp}(1), X, Y$ independent

$$f(x, y) = \begin{cases} e^{-(x+y)}, & x, y > 0 \\ 0, & \text{otherwise} \end{cases}$$

$$F_z(z) = P\left(\frac{X}{Y} < z\right) = \iint_{x > 0, y > 0, \frac{x}{y} < z} e^{-(x+y)} dx dy = \int_0^\infty \int_0^{yz} e^{-(x+y)} dx dy = 1 - \frac{1}{1+z}$$

$$\Rightarrow f_z(z) = F'_z(z) = \begin{cases} (1+z)^{-2}, & z > 0 \\ 0, & \text{otherwise} \end{cases}$$

3.4 Multivariate Normal Distribution

Bivariate Normal Distribution

$(X, Y) \Rightarrow \mu_x, \mu_y, \sigma_x^2, \sigma_y^2, \rho$ (Correlation coefficient ρ)

$$f(x, y) = \frac{1}{2\pi\sigma_x\sigma_y\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)}\left[\frac{(x-\mu_x)^2}{\sigma_x^2} - 2\rho\frac{(x-\mu_x)(y-\mu_y)}{\sigma_x\sigma_y} + \frac{(y-\mu_y)^2}{\sigma_y^2}\right]\right)$$

$$\downarrow$$

$$\begin{pmatrix} X \\ Y \end{pmatrix} \sim N\left(\begin{pmatrix} \mu_x \\ \mu_y \end{pmatrix}, \begin{pmatrix} \sigma_x^2 & \rho\sigma_x\sigma_y \\ \rho\sigma_x\sigma_y & \sigma_y^2 \end{pmatrix}\right) = N(\mu, \Sigma)$$

μ is the mean vector 均值向量

Σ is the variance-covariance matrix 协方差矩阵

Specially if $\mu_x = \mu_y = 0, \sigma_x = \sigma_y = 1, \rho > 0 \Rightarrow$ standard bivariate distribution $N(0, I)$

Identity matrix

* If X, Y follow a bivariate distribution,

Uncorrelated $\Rightarrow$ Independent

* Marginal distributions are normal distributions

$$X \sim N(\mu_x, \sigma_x^2), Y \sim N(\mu_y, \sigma_y^2), \rho_{XY} = \rho$$

Conditional distributions are normal distributions

$$X|Y=y \sim N\left(\mu_x + \frac{\rho\sigma_x}{\sigma_y}(y-\mu_y), (1-\rho^2)\sigma_x^2\right)$$

$$Y|X=x \sim N\left(\mu_y + \frac{\rho\sigma_y}{\sigma_x}(x-\mu_x), (1-\rho^2)\sigma_y^2\right)$$

Linear Combinations still follow normal distribution

$$C_1 X + C_2 Y \sim N(C_1 \mu_x + C_2 \mu_y, C_1^2 \sigma_x^2 + 2\rho C_1 C_2 \sigma_x \sigma_y + C_2^2 \sigma_y^2)$$

Matrix $A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$

$$A \begin{pmatrix} X \\ Y \end{pmatrix} = \begin{pmatrix} a_{11}X + a_{12}Y \\ a_{21}X + a_{22}Y \end{pmatrix} \sim N(A\mu, A\Sigma A^T)$$

Let $U_{11}, U_{12}, \dots, U_{n1}, U_{n2}$ be $2n$ numbers from $U[0,1]$

$$Z_{2i-1} = \sqrt{2 \ln(1/U_{1i})} \cos(2\pi U_{1i}) \quad i=1, 2, \dots, n$$

$$Z_{2i} = \sqrt{2 \ln(1/U_{2i})} \sin(2\pi U_{2i})$$

$Z_1, Z_2, \dots, Z_{2n} \Rightarrow$ generate from $N(0,1)$

$Y_i = \mu + Z_i \sigma \Rightarrow$ generate from $N(\mu, \sigma^2)$

Rejection sampling

give the target PDF $f(x)$ distribution target distribution

$\therefore$ difficult to sample from the target distribution

$\therefore$ Sample from a distribution with PDF $g(x)$ proposal distribution

Number from $g(x)$ either accepted or rejected
 $\downarrow$
 sample $f(x)$

Let (X,Y) have uniform distribution over region D

$D = \{(x,y) : 0 \leq y \leq f(x), a \leq x \leq b\}$ for some PDF $f(x)$

$X \sim f(x)$

* $g(x)$ has same support as $f(x)$

$\exists c > 1, \forall x f(x) \leq c g(x)$

1. Sample X_i from $g(x)$

2. Sample U_i from $U[0,1]$

reject if $U_i > \frac{f(X_i)}{c g(X_i)}$

Acceptance Probability of Rejection sampling

$$P(U \leq \frac{f(x)}{c g(x)}) = \int_{-\infty}^{\infty} P(U \leq \frac{f(x)}{c g(x)} | x=x) g(x) dx = \frac{1}{c}$$

If possible, $c = \sup \frac{f(x)}{g(x)}$
上确界

Accepted numbers from the rejection sampling follows the target distribution $f(x)$

$(X_{(1)}, X_{(2)}, \dots, X_{(n)})$ 顺序统计量

Q_p 样本 p 分位数

$$Q_p = \begin{cases} X_{(\lfloor np \rfloor + 1)} & , np \text{ is not an integer} \\ \frac{X_{(np)} + X_{(np+1)}}{2} & , np \text{ is an integer} \end{cases}$$

$Q_{0.5}$ 样本中位数 $Q_{0.25}$ 样本第一四分位数

5.2 Point estimation

Properties of point estimation

Let $X_1, \dots, X_n$ be a simple random sample from

$X \sim f(x; \theta_1, \dots, \theta_k)$ $\theta_1, \dots, \theta_k$ are unknown parameters

$\hat{\theta}_L(X_1, \dots, X_n)$ is an estimator (估计量) of θ_L

$\hat{\theta}_L(x_1, \dots, x_n)$ is an estimate (估计值) of θ_L

Unbiasedness

Θ is the parameter space 参数空间 of $\theta = (\theta_1, \dots, \theta_k)$

If for $\forall \theta \in \Theta$, $E(\hat{\theta}_L) = \theta_L$

the $\hat{\theta}_L$ is an unbiased estimator 无偏估计量 of θ_L

or it is a biased estimator 有偏估计量 of θ_L

$E(\hat{\theta}_L) - \theta_L$ is the bias of the estimator 渐近无偏估计量

if bias $\neq 0$, but $\rightarrow 0$ when $n \rightarrow \infty \rightarrow$ asymptotic unbiased estimator

$$f_{\max}(x) = n f(x) [F(x)]^{n-1} = n \cdot \frac{1}{\theta} \cdot \frac{x^{n-1}}{\theta^{n-1}} = \frac{n}{\theta^n} x^{n-1}$$

$$f_{\min}(x) = n f(x) [1-F(x)]^{n-1} = n \cdot \frac{1}{\theta} \cdot [1 - \frac{x}{\theta}]^{n-1} = \frac{n}{\theta} (1 - \frac{x}{\theta})^{n-1}$$

Figure out size.

$$E(X_{(n)}) \begin{cases} \xrightarrow{\max} E(X) = \int_{-\infty}^{\infty} x f(x) dx = \int_0^{\theta} \dots dx \\ \xrightarrow{\min} \\ \xrightarrow{k\text{-th smallest}} \end{cases}$$

$$f_{X_k}(x) = \frac{n!}{(k-1)!(n-k)!} \left(\frac{x}{\theta}\right)^{k-1} (1 - \frac{x}{\theta})^{n-k} \frac{1}{\theta}, 0 \leq x \leq \theta$$

$$\text{Var}(X_{(n)}) = E(X_{(n)}^2) - E^2(X_{(n)})$$

$$E(X_{(n)}^2) = \int_{-\infty}^{\infty} x^2 f(x) dx = \int_0^{\theta} \dots dx$$

Consistency

If for $\forall \theta \in \Theta$ and $\forall \varepsilon > 0$

$$\lim_{n \rightarrow \infty} P(|\hat{\theta}_L - \theta_L| > \varepsilon) = 0 \Rightarrow \hat{\theta}_L \xrightarrow{P} \theta_L, \text{ as } n \rightarrow \infty$$

If $\hat{\theta}$ is an 渐近无偏估计量 & $\text{Var}(\hat{\theta}) \rightarrow 0$, as $n \rightarrow \infty$

$\Rightarrow \hat{\theta}$ is a consistent estimator of θ

Chebyshev's 不等式: $X, E(X) = \mu, \sigma^2 = \text{Var}(X), P(|X - \mu| \leq k\sigma) \leq \frac{1}{k^2}$

Method of moments 矩估计法

Moments: Let $X_1, \dots, X_n$ be sample from $X \sim f(x; \theta_1, \dots, \theta_k)$

总体 k 阶矩 $\mu_k = E(X^k)$ 总体矩中心矩 $\tilde{\mu}_k = E[(X - \mu_1)^k]$

样本 k 阶矩 $M_k = \frac{1}{n} \sum_{i=1}^n X_i^k$ 样本矩中心矩 $\tilde{M}_k = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^k$

$$(\tilde{M}_2 = M_2 - M_1^2)$$

总体矩 = 参数的函数 = 样本矩

$$\begin{cases} \mu_1 = h_1(\theta_1, \dots, \theta_k) = M_1 \\ \mu_k = h_k(\theta_1, \dots, \theta_k) = M_k \end{cases} \Rightarrow \begin{cases} \theta_1 = g_1(M_1, \dots, M_k) \\ \theta_k = g_k(M_1, \dots, M_k) \end{cases}$$

for $l=1, \dots, k$, $\hat{\theta}_L = g_L(M_1, \dots, M_k) = \tilde{g}_L(X_1, \dots, X_n)$ 矩估计

$$\hat{\mu}_1 = M_1 = \bar{X} = E(X) \quad \hat{\sigma}^2 = \tilde{M}_2 = \frac{(n-1)}{n} S^2$$

$$\hat{\mu}_2 = E(X^2) = \text{Var}(X) + E^2(X)$$

Chapter 5 Basic Concepts in Stats

5.1 Population, Sample, Parameters, Stats

population 总体 parameter 参数 sample 样本

parametric statistical inference 参数估计推断

quantitative index 定量指标 X

Assume $X \sim F(x; \theta)$ or $X \sim f(x; \theta)$ e.g. $X \sim N(\mu, \sigma^2)$, $\theta = (\mu, \sigma^2)$

sample of size n $X_1, \dots, X_n$

Each individual equal probability of being chosen

$X_1, \dots, X_n$ i.i.d X

If $X_1, \dots, X_n$ is a sample from $X \sim f(x; \theta)$

$g(X_1, \dots, X_n)$ is an n -variate function

$T = g(X_1, \dots, X_n)$ is statistic (估计量) if $g(\cdot)$ doesn't have any unknown parameter.

Commonly used:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i \text{ 样本均值} \quad S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 \text{ 样本方差}$$

Method of Maximum Likelihood 最大似然估计

$$L(\theta; x_1, \dots, x_n) = \prod_{i=1}^n f(x_i; \theta), \theta \in \Theta$$

If $\exists \hat{\theta} = \hat{\theta}(x_1, \dots, x_n)$ that

$$L(\hat{\theta}) = \max_{\theta \in \Theta} L(\theta; x_1, \dots, x_n)$$

$\hat{\theta}(x_1, \dots, x_n)$ is 极大似然估计值

$\hat{\theta}(x_1, \dots, x_n)$ is 极大似然估计量

$$\ell(\theta; x) = \log L(\theta; x) = \sum_{i=1}^n \log f(x_i; \theta)$$

$$\frac{d\ell(\theta; x)}{d\theta} = 0 \Rightarrow \hat{\theta}_{MLE} \Rightarrow \text{parameter space restriction}$$

when $\frac{d\ell(\theta; x)}{d\theta}$ can't be 0, 找0使L最大

Confidence Interval 置信区间

$$\forall \alpha \in (0, 1), \exists \hat{\theta}_{L1}(x_1, \dots, x_n), \hat{\theta}_{L2}(x_1, \dots, x_n)$$

$$P_{\theta}(\hat{\theta}_{L1} < \theta < \hat{\theta}_{L2}) \geq 1 - \alpha, \forall \theta \in \Theta$$

$\Rightarrow (\hat{\theta}_{L1}, \hat{\theta}_{L2})$ is a confidence interval of θ_L with level $1 - \alpha$,
or $100(1 - \alpha)\%$ confidence interval

$$1 - \alpha = P(\mu - C_2 < \bar{X} < \mu + C_1) = P\left(\frac{-C_2}{\sigma/\sqrt{n}} < \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < \frac{C_1}{\sigma/\sqrt{n}}\right)$$

$$= P\left(1 - \frac{C_2}{\sigma} < Z < \frac{C_1}{\sigma}\right)$$

$$C_1 = C_2 = Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \quad Z_{\alpha/2} \text{ 上 } \alpha/2 \text{ 分位点}$$

$$(\bar{X} - Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{X} + Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}) \text{ is CI of } 100(1 - \alpha)\%$$

General Confidence Interval Construction

$x_1, \dots, x_n$ from $X \sim f(x; \theta_1, \dots, \theta_k)$

$\hat{\theta}_L(x_1, \dots, x_n)$ is unbiased estimator of θ_L , and the standard deviation of $\hat{\theta}_L$ is $\sigma(\hat{\theta}_L) = SD(\hat{\theta}_L)$ Standard Error

If $\hat{\theta}_L$ exactly follows $N(\theta_L, \sigma^2(\hat{\theta}_L))$, and $\sigma^2(\hat{\theta}_L)$ doesn't depend on $\forall$ unknown parameter, then an EXACT $100(1 - \alpha)\%$ CI of θ_L

$$\hat{\theta}_L \pm Z_{\alpha/2} \sigma(\hat{\theta}_L) = (\hat{\theta}_L - Z_{\alpha/2} \sigma(\hat{\theta}_L), \hat{\theta}_L + Z_{\alpha/2} \sigma(\hat{\theta}_L))$$

If $\hat{\theta}_L \stackrel{\text{approx.}}{\sim} N(\theta_L, \sigma^2(\hat{\theta}_L))$ or $\sigma^2(\hat{\theta}_L)$ depends on unknown parameters and $\hat{\sigma}^2(\hat{\theta}_L)$ is a consistent estimator of $\sigma^2(\hat{\theta}_L)$ then a large sample $100(1 - \alpha)\%$ CI of θ_L is

$$\hat{\theta}_L \pm Z_{\alpha/2} \hat{\sigma}(\hat{\theta}_L) = (\hat{\theta}_L - Z_{\alpha/2} \hat{\sigma}(\hat{\theta}_L), \hat{\theta}_L + Z_{\alpha/2} \hat{\sigma}(\hat{\theta}_L))$$

Use CLT, $\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$

$$\bar{X} - \bar{Y} \stackrel{\text{approx.}}{\sim} N\left(\mu_X - \mu_Y, \frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}\right) \quad \text{independent } X, Y$$

Chapter 6 Hypothesis Testing

6.1 Overview and basic ideas

Verify hypothesis

Null hypothesis: 零假设 H_0 , No effect

Alternative hypothesis: 备择假设 H_1 , presence of effect

Rejection region: 拒绝域 R : when $X \in R$, we reject H_0 .

Two Types of errors

Type I: reject H_0 when H_0 is true

$$\text{错误率 } \alpha = P\{\text{reject } H_0 \mid H_0 \text{ is true}\} = P\{X \in R \mid H_0 \text{ is true}\}$$

Type II: not reject H_0 when H_0 is false

$$\text{错误率 } \beta = P\{\text{not reject } H_0 \mid H_0 \text{ is false}\} = P\{X \notin R \mid H_0 \text{ is false}\}$$

H_1 is true

Significance test 显著性检验

minimizing β that $\alpha < \alpha$ given level

Let $X = (X_1, \dots, X_n)$ from $X \sim f(x; \theta_1, \dots, \theta_k)$

For testing $H_0: \theta_L \in \Theta_{L0} \leftrightarrow H_1: \theta_L \in \Theta_{L1}, \forall \alpha \in (0, 1)$

$$P_{\theta}(X \in R) \leq \alpha, \forall \theta \in \Theta \text{ with } \theta_L \in \Theta_{L0}$$

显著性水平为 α 的显著性检验 $H_0: \mu_X \leq 8$

simple null hypothesis: 简单原假设 $H_0^*: \mu_X = 8$

P value = $P(\text{observed or extreme data} \mid H_0 \text{ is true})$

reject H_0 if p-value is small

Reject H_0 if p-value $< \alpha$, otherwise fail to reject H_0

检验统计量 $T = \frac{\bar{X} - \mu}{\sigma_X / \sqrt{n}}$ or large sample $\frac{\bar{X} - \mu}{\sigma_X / \sqrt{n}} \sim N(0, 1)$ under H_0

p-value = $P(T \text{ more extreme} \mid H_0 \text{ is true})$

power: $P(\text{reject } H_0 \mid H_1 \text{ true})$

$$\text{power} = P(X \in R \mid H_1 \text{ is true}) = 1 - \beta$$

$\Rightarrow$ find minimum sample size

6.2 General Process

1. Formulate H_0, H_1

2. Choose a proper test statistic T

Derive distribution of T under H_0 is true

3. Compute the observed value of T

4. obtain p-value and compare it with α

we know if reject H_0 at significance level α

H_0 : something false (maybe)

H_1 : something you want to prove